RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

MINI PROJECT REPORT

Submitted by

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AI23331 – FUNDAMENTALS OF MACHINE LEARING

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BONAFIDE CERTIFICATE

Certified that this project report titled **"Placement Prediction Using Machine Learning"** is the bonafide work of **YELESWARAM BHARGAV (231701063)** who carried out the project work for the course **AI23331 - Fundamentals of Machine Learning** under my supervision.

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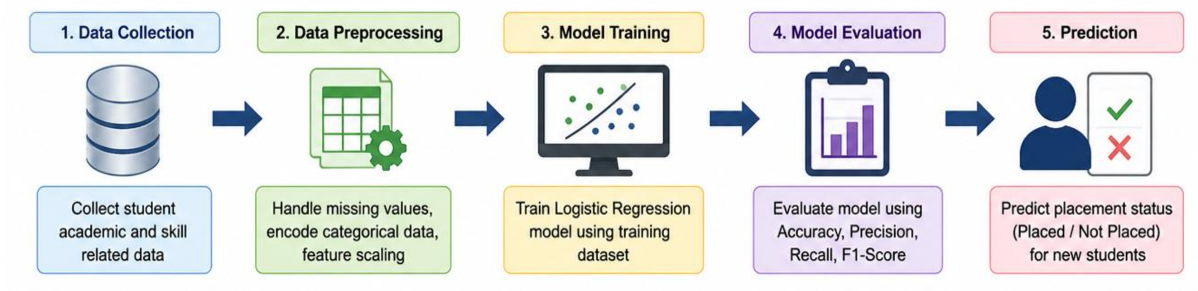
CHAPTER 1: INTRODUCTION

1.1 Introduction

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn from data and make predictions. In the education sector, machine learning can be used to predict student placement outcomes based on academic and skill-related factors.

This project, **“Placement Prediction Using Machine Learning”**, predicts whether a student is likely to be placed or not using features such as **CGPA, internships, projects, aptitude test score, soft skills, certifications, and academic marks**. The system uses the **Logistic Regression algorithm** to analyze student data and generate placement predictions. This helps students understand their placement readiness and improve their performance. Based on the project files, the model uses student-related features and logistic regression for prediction.

Image 1.1: Machine Learning in Placement Prediction Workflow



1.2 Problem Statement

Many students face uncertainty regarding placement opportunities due to differences in academic performance, technical skills, and extracurricular involvement. Traditional placement evaluation methods are time-consuming and may not accurately predict student placement outcomes. Therefore, there is a need for an intelligent system that can analyze student-related factors and predict placement status effectively.

The objective of this project is to develop a machine learning-based placement prediction system that analyzes student academic and skill-related data to predict whether a student will be placed or not.

1.3 Objective

The main objectives of the project are:

- To analyze student placement data.
- To preprocess and prepare data for machine learning.
- To implement the Logistic Regression algorithm for placement prediction.
- To predict student placement status based on academic and skill-related features.
- To evaluate model performance using accuracy, precision, recall, and F1-score.
- To help students understand placement requirements and improve their skills.

1.4 Existing System

In the existing system, placement prediction is mainly based on manual analysis performed by faculty members or placement officers. Student performance is evaluated using academic marks and aptitude results, which may not provide accurate predictions. Manual methods require more time and may lead to inconsistent decision-making.

Limitations of Existing System

- Time-consuming process
- Less accurate predictions
- Human errors in evaluation
- Limited use of multiple student performance factors

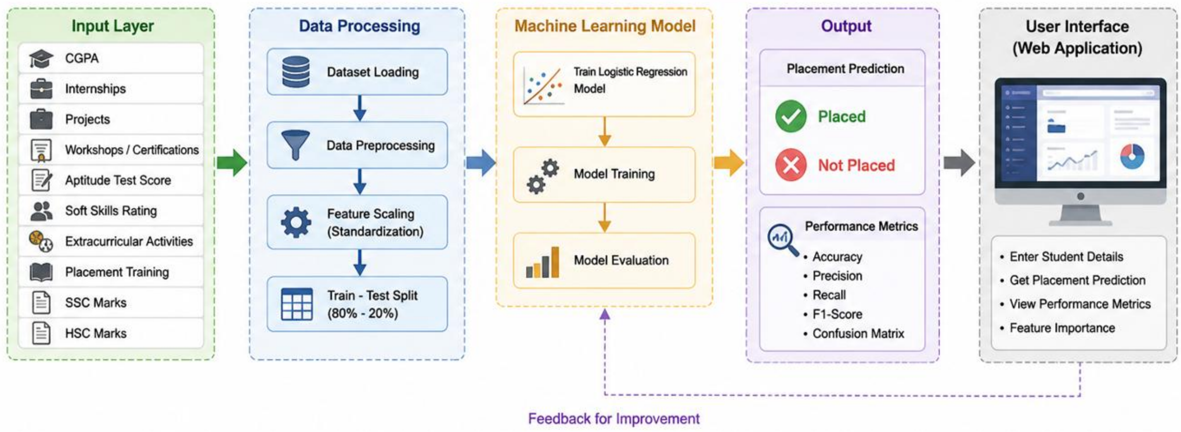
1.5 Proposed System

The proposed system uses a **Machine Learning-based Placement Prediction Model** to predict whether a student will be placed or not. The system takes multiple student-related inputs such as academic performance, internships, projects, aptitude scores, and soft skills to generate placement predictions. The project uses the **Logistic Regression algorithm**, data preprocessing, feature scaling, and model evaluation techniques for prediction. The model uses training and testing datasets to improve prediction accuracy and reliability. The implementation includes dataset loading, feature preprocessing, model training, and prediction generation.

Advantages of Proposed System

- Fast and automated prediction
- Better accuracy in placement analysis
- Uses multiple performance factors
- Helps students improve placement readiness
- Reduces manual effort in evaluation

Image 1.2: Proposed System Architecture



CHAPTER 2: SYSTEM DESIGN

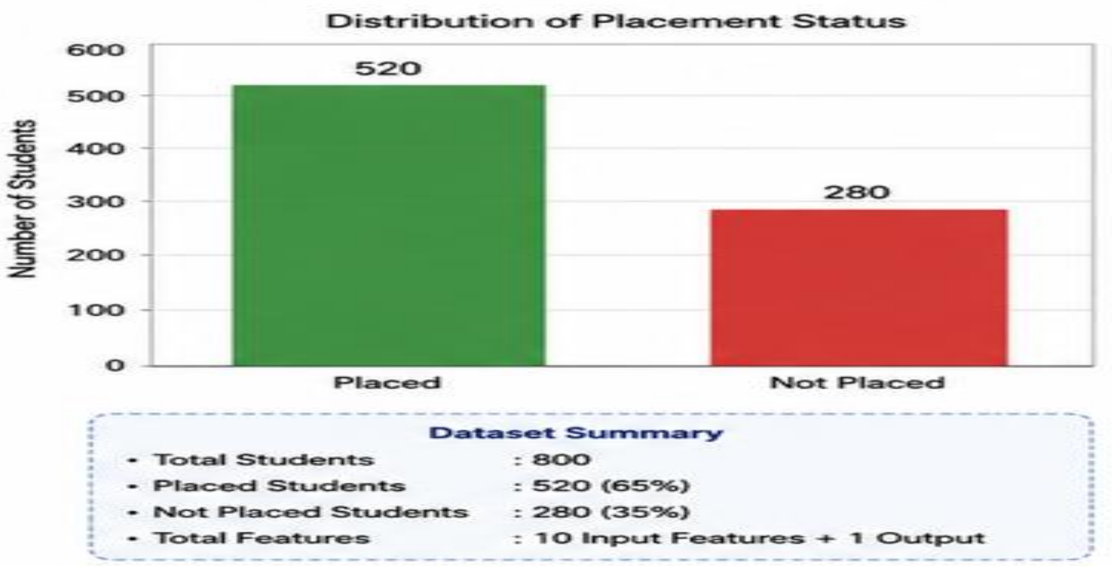
2.1 Dataset Description

The dataset used in this project contains student academic and skill-related information for placement prediction. The data is collected in CSV format and includes various features that influence placement opportunities. The dataset is used to train and test the machine learning model for accurate prediction.

Table 2.1: Dataset Features

S.No	Feature Name	Description
1	CGPA	Student academic performance
2	Internships	Number of internships completed
3	Projects	Number of projects completed
4	Workshops/Certifications	Certifications and workshops attended
5	Aptitude Test Score	Aptitude performance score
6	Soft Skills Rating	Communication and soft skill level
7	Extracurricular Activities	Participation in activities
8	Placement Training	Placement preparation training
9	SSC Marks	Secondary school marks
10	HSC Marks	Higher secondary marks
11	Placement Status	Output (Placed/Not Placed)

Image 2.1: Dataset Overview Chart



2.2 Data Preprocessing

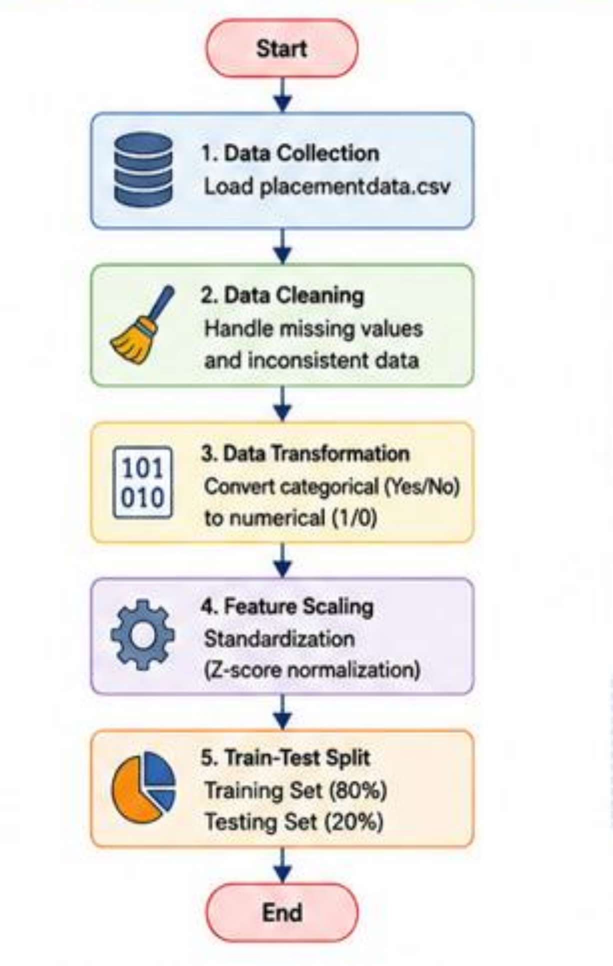
Data preprocessing is an important step in machine learning. It helps in preparing raw data before training the model. In this project, preprocessing includes converting data into numerical values and preparing features for model training.

The dataset is divided into **training data (80%)** and **testing data (20%)** for model evaluation. Feature scaling is also applied to improve model performance and prediction accuracy. The project converts binary values such as extracurricular activities and placement training into numerical form for processing.

Steps in Data Preprocessing

- Data collection from dataset
- Data cleaning and formatting
- Feature conversion into numerical values
- Feature scaling (standardization)
- Train-test split (80:20)

Image 2.2: Data Preprocessing Flowcha



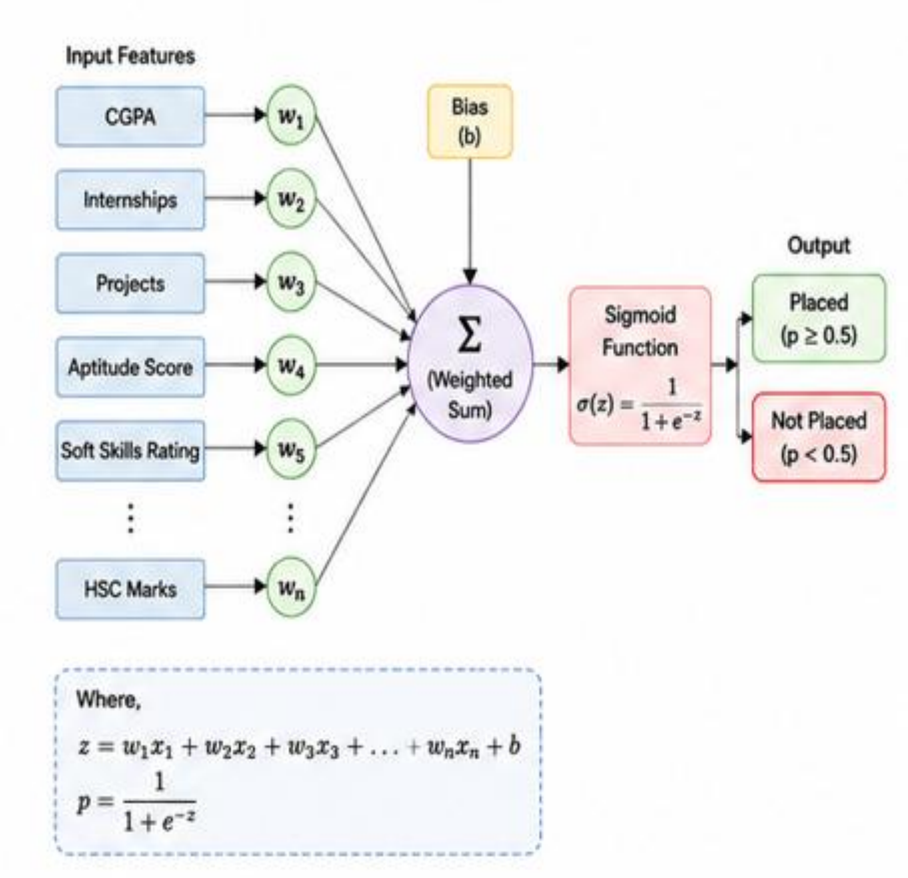
2.3 Algorithm Used – Logistic Regression

This project uses the **Logistic Regression algorithm** for placement prediction. Logistic Regression is a supervised machine learning algorithm used for classification problems. It predicts whether a student will be **Placed** or **Not Placed** based on input features.

The algorithm applies the sigmoid function to calculate prediction probability and classify student placement status. Logistic Regression is selected because it is simple, efficient, and suitable for binary classification problems. The model is trained using student placement data and tested for prediction accuracy.

Advantages of Logistic Regression

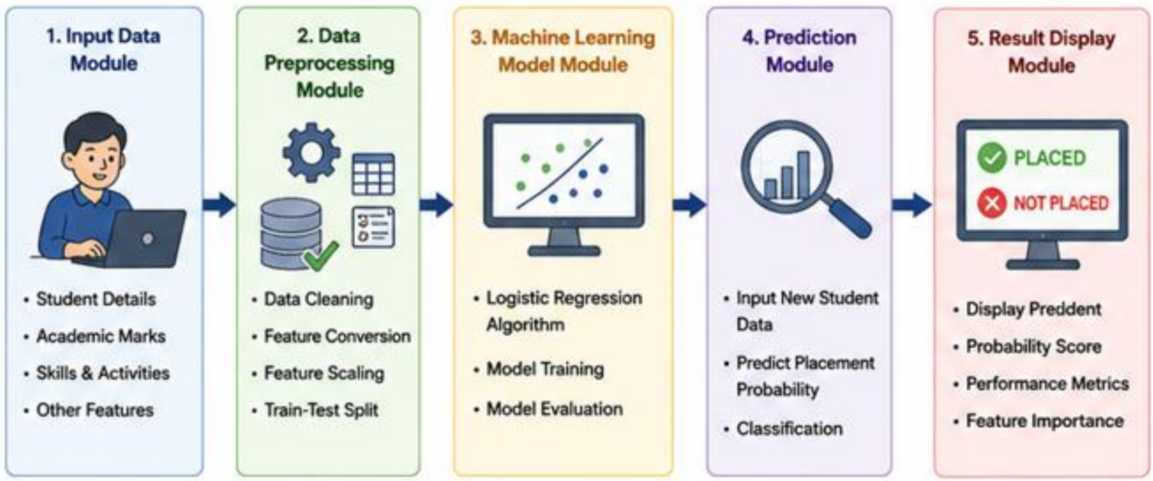
- Simple and easy to implement
- Suitable for binary classification
- Provides fast predictions
- Works effectively with structured datasets



2.4 System Architecture

The system architecture describes the working flow of the placement prediction system. Student input data such as academic scores, projects, and internships are collected and processed. The machine learning model analyzes these inputs and predicts placement status.

Image 2.4: System Architecture Diagram



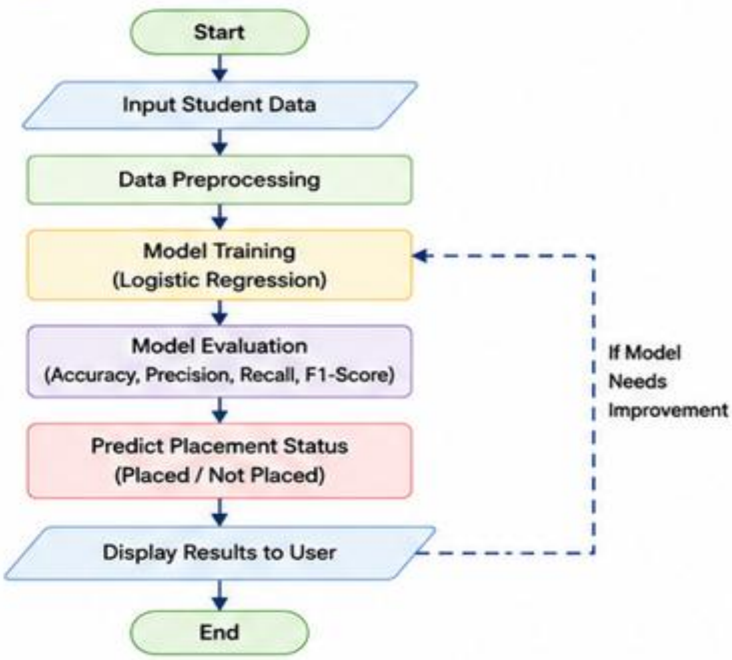
2.5 Flowchart

The flowchart explains the overall working process of the placement prediction system. It starts with student data collection, followed by preprocessing, model training, prediction, and result generation.

Flow of the System:

Student Data → Preprocessing → Model Training → Prediction → Result

Image 2.5: Placement Prediction Flowchart



CHAPTER 3: IMPLEMENTATION

3.1 Software and Hardware Requirements

The implementation of the placement prediction system requires both software and hardware components for model training, testing, and prediction. Python is used for developing the machine learning model, while CSV files are used for storing student placement data.

Table 3.1: Software Requirements

S.No	Software	Purpose
1	Python	Programming language
2	VS Code / Any IDE	Code development
3	CSV Dataset	Data storage
4	Web Browser	Viewing results

Table 3.2: Hardware Requirements

S.No	Hardware	Specification
1	Processor	Intel i3 or above
2	RAM	4 GB or above
3	Storage	**100 GB minimum**
4	Operating System	Windows/Linux

3.2 Modules Description

The placement prediction system is divided into different modules to perform specific operations efficiently.

1. Dataset Loading Module

This module loads the placement dataset from the CSV file and converts it into a suitable format for processing. The uploaded project uses `placementdata.csv` for training and testing the model.

2. Data Preprocessing Module

This module converts raw student data into numerical format, handles binary values, and prepares the dataset for training. Feature scaling is applied to improve model performance.

3. Machine Learning Module

The machine learning module trains the **Logistic Regression algorithm** using training data. The sigmoid function is used to predict placement probability.

4. Prediction Module

This module accepts student input details and predicts whether the student will be **Placed** or **Not Placed**. It also calculates confidence and prediction probability.

5. Result Display Module

The final module displays prediction results and performance metrics such as accuracy, precision, recall, and F1-score to the user.

3.3 Model Training and Prediction

In this project, the Logistic Regression algorithm is used to train the placement prediction model. The dataset is divided into 80% training data and 20% testing data. Student details such as CGPA, internships, projects, aptitude score, soft skills, and academic marks are used to predict whether a student will be Placed or Not Placed. The model is trained using historical placement data and evaluated for accuracy.

Steps in Model Training

1. Load dataset
2. Preprocess data
3. Split dataset into train and test data
4. Train Logistic Regression model
5. Predict placement status

Dataset Splitting

```
def predict_placement(payload):
    probability = predict_probability(
        scaled_row,
        MODEL_BUNDLE["model"]
    )
    return {
        "prediction":
            "Placed" if probability >= 0.5
            else "NotPlaced"
    }
```

Logistic Regression Training

```
def train_logistic_regression(
    features, labels, learning_rate=0.08, epochs=1600
):
    weights = [0.0] * len(features[0])
    bias = 0.0

    for _ in range(epochs):
        for row, actual in zip(features, labels):
            linear = bias + sum(
                weight * value
                for weight, value in zip(weights, row)
            )
```

Placement Prediction

```
def split_dataset(rows):
    train_size = int(len(rows) * 0.8)
    return rows[:train_size], rows[train_size:]
```

CHAPTER 4: RESULTS AND DISCUSSION

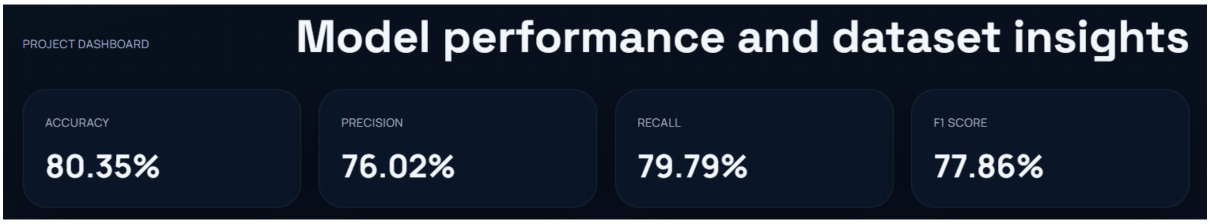
4.1 Performance Metrics

The performance of the placement prediction model is evaluated using different metrics such as **Accuracy, Precision, Recall, and F1-Score**. These metrics help in measuring the effectiveness of the Logistic Regression model in predicting placement status.

Table 4.1: Performance Metrics

Metric	Description
Accuracy	Measures overall prediction correctness
Precision	Measures correct positive predictions
Recall	Measures ability to identify placed students
F1-Score	Balance between precision and recall

The model performance is evaluated after testing the trained model on student placement data. The project calculates these metrics to improve prediction accuracy.

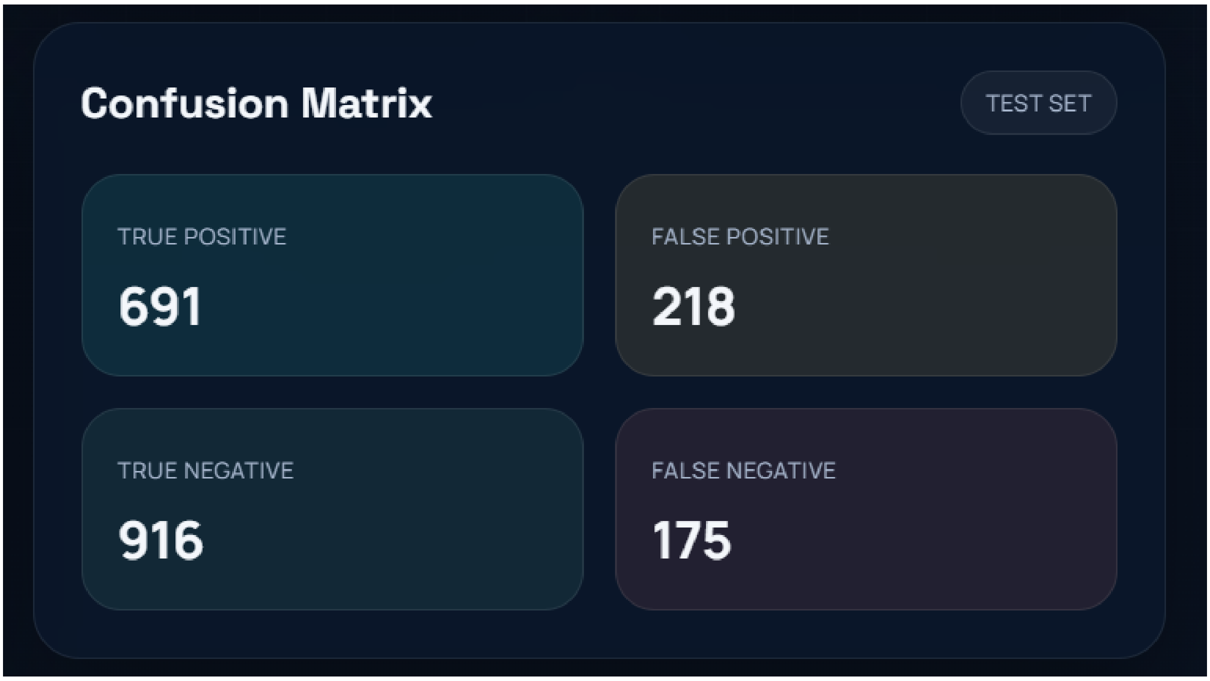


4.2 Confusion Matrix

A confusion matrix is used to evaluate classification performance by comparing actual and predicted values. It shows correctly and incorrectly predicted placement results.

Actual / Predicted	Placed	Not Placed
Placed	True Positive (TP)	False Negative (FN)
Not Placed	False Positive (FP)	True Negative (TN)

The confusion matrix helps in understanding how accurately the model predicts placement status and identifies prediction errors. The project implementation includes confusion matrix calculation for evaluation.



4.3 Prediction Results

The system predicts whether a student will be **Placed** or **Not Placed** based on input features such as **CGPA, internships, projects, aptitude score, soft skills, certifications, SSC marks, and HSC marks**.

Sample Prediction Table

CGPA	Internships	Projects	Aptitude Score	Prediction
8.5	2	3	80	Placed
6.8	0	1	55	Not Placed
9.0	3	4	90	Placed


Top factors influencing placement
FEATURE IMPORTANCE
Aptitude Test Score
100%
Placement Training
75%
Extracurricular Activities
63%
SSC Marks
61%
Soft Skills Rating
48%
HSC Marks
47%
Projects
45%
CGPA
44%
Workshops & Certifications
22%
Internships
3%

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project, **“Placement Prediction Using Machine Learning”**, was developed to predict whether a student is likely to be **Placed** or **Not Placed** based on academic and skill-related factors. The system uses the **Logistic Regression algorithm** to analyze student data such as **CGPA, internships, projects, aptitude score, soft skills, certifications, SSC marks, and HSC marks**.

The model was trained and tested using placement data and evaluated using performance metrics such as **accuracy, precision, recall, and F1-score**. The project provides a simple and effective solution for predicting student placement outcomes and helps students improve their placement readiness. The implementation includes model training, prediction, and performance evaluation.

5.2 Future Scope

The following improvements can be added to the project in the future:

- Use advanced machine learning algorithms for better accuracy.
- Increase dataset size for improved prediction performance.
- Add more student-related features for analysis.
- Develop a mobile-friendly application interface.
- Integrate real-time student performance tracking.

This project can be further enhanced to provide more accurate and intelligent placement prediction systems for educational institutions.

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2. Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media.
3. Python Software Foundation. *Python Programming Language*. Available: https://www.python.org/
4. Microsoft. *Visual Studio Code*. Available: https://code.visualstudio.com/
5. Student Placement Dataset (placementdata.csv) used for training and testing.
6. Project source code files:
    - data_loader.py – Dataset loading and preprocessing
    - ml_model.py – Logistic Regression model training
    - model_service.py – Prediction and evaluation
    - server.py – Backend server implementation